

1.

C++ code	Result
<pre> #include<iostream> using namespace std; int main() { const char* err = "Can't get the memory!!"; int block_size; cout << "Input the size of the array : "; cin >> block_size; int* p = new(std::nothrow) int[block_size]; try { if (p == NULL) throw err; } catch (const char* s) { cerr << s << endl; } cout << endl; cin.clear(); p = new(std::nothrow)int[100]; cout << "The size of the array has just been set to 100." << endl; try { if (p == NULL) throw err; for (int n = 0; n < 100; n++) p[n] = n; int n; cout << "Input a integer n from 0 - 99, then output array[n]" << endl; cin >> n; if (n < 0) throw "underflow!!"; else if (n >= 100) throw "overflow!!"; else cout << "p[" << n << "] = " << p[n] << endl; } catch (const char* s) { cerr << s << endl;; } cin.clear(); cout << endl; try { delete[]p; p = NULL; if (p == NULL) throw "A null pointer can't be delete."; } catch (const char *s) { cout << s << endl; } } </pre>	Input the size of the array : 1000000000 Can't get the memory!! The size of the array has just been set to 100. Input a integer n from 0 - 99, then output array[n] 101 overflow!! A null pointer can't be delete.

2.

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<pre>#include<iostream> #include<algorithm> #include<functional> #include<vector> #include<iterator> #include<string> #include<iomanip> using namespace std; class student { private: string name; int sno; int score; public: student(string s = " ", int no = 0, int sco = 0){ try { if (no < 0 sco < 0) throw "Input wrong!!"; name = s; sno = no; score = sco; } catch (const char* s) { cerr << s << endl; } } student(const student& s) : name(s.name), sno(s.sno), score(s.score) {} ~student() {} void Set(char* s, int no, int sco) { try { if (no < 0 sco < 0) throw "Input wrong!!"; name = s; sno = no; score = sco; } catch (const char* s) { cerr << s << endl; } } const string& Getname() const{ return name; } const int Getsno() const { return sno; } const int Getscore() const{ return score; } void Display() { cout << "Name : " << setw(12) << setiosflags(ios_base::left) << name << " student no." << setw(6) << sno << " Score : " << setw(6) << score << endl; } student& operator = (const student& stu) { name = stu.name; sno = stu.sno; score = stu.score; return *this; } }</pre>	Input the number of a class (greater than 0 and less than 16) : 16 <table><tr><td>Name : Devil</td><td>student no.6</td><td>Score : 98</td></tr><tr><td>Name : Geogria</td><td>student no.8</td><td>Score : 86</td></tr><tr><td>Name : Maria</td><td>student no.14</td><td>Score : 82</td></tr><tr><td>Name : Felix</td><td>student no.7</td><td>Score : 66</td></tr><tr><td>Name : David</td><td>student no.5</td><td>Score : 56</td></tr><tr><td>Name : Jacqueline</td><td>student no.11</td><td>Score : 54</td></tr><tr><td>Name : Honest</td><td>student no.9</td><td>Score : 51</td></tr><tr><td>Name : Lagrange</td><td>student no.13</td><td>Score : 43</td></tr><tr><td>Name : Apple</td><td>student no.1</td><td>Score : 39</td></tr><tr><td>Name : Cat</td><td>student no.3</td><td>Score : 39</td></tr><tr><td>Name : Corona</td><td>student no.4</td><td>Score : 38</td></tr><tr><td>Name : Marina</td><td>student no.15</td><td>Score : 38</td></tr><tr><td>Name : Notage</td><td>student no.16</td><td>Score : 22</td></tr><tr><td>Name : Banana</td><td>student no.2</td><td>Score : 20</td></tr><tr><td>Name : Isabella</td><td>student no.10</td><td>Score : 13</td></tr><tr><td>Name : Kasin</td><td>student no.12</td><td>Score : 1</td></tr></table>	Name : Devil	student no.6	Score : 98	Name : Geogria	student no.8	Score : 86	Name : Maria	student no.14	Score : 82	Name : Felix	student no.7	Score : 66	Name : David	student no.5	Score : 56	Name : Jacqueline	student no.11	Score : 54	Name : Honest	student no.9	Score : 51	Name : Lagrange	student no.13	Score : 43	Name : Apple	student no.1	Score : 39	Name : Cat	student no.3	Score : 39	Name : Corona	student no.4	Score : 38	Name : Marina	student no.15	Score : 38	Name : Notage	student no.16	Score : 22	Name : Banana	student no.2	Score : 20	Name : Isabella	student no.10	Score : 13	Name : Kasin	student no.12	Score : 1
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    }
};

bool compare(const student& s1, const student& s2) {
    if (s1.Getscore() > s2.Getscore())
        return true;
    else if (s1.Getscore() == s2.Getscore()) {
        if (s1.Getname() < s2.Getname())
            return true;
        else if ((s1.Getname() == s2.Getname()) &&
(s1.Get sno() < s2.Get sno()))
            return true;
        else
            return false;
    }
    else
        return false;
}

int main() {
    int nc;
    char NAME[16][20] = { "Apple", "Banana", "Cat",
"Corona", "David", "Devil", "Felix", "Geogria",
"Honest", "Isabella", "Jacqueline", "Kasin",
"Lagrange", "Maria", "Marina", "Notage" };
    cout << "Input the number of a class (greater
than 0 and less than 16) : ";
    cin >> nc;
    student* stu = new student[nc*sizeof(student)];
    srand(0);
    for (int n = 0; n < nc; n++)
        (stu + n)->Set(NAME[n], n + 1, rand() %
100 + 1);
    vector<student> CLASS;
    for (int n = 0; n < nc; n++)
        CLASS.push_back(stu[n]);
    sort(CLASS.begin(), CLASS.end(), compare);
    for (int n = 0; n < nc; n++)
        CLASS[n].Display();
    delete[] stu;
}

```